Accessible Emotion-Aware E-Learning Platform for Deaf Students Studying O/L Information Technology Using Facial Expression Recognition and an NLP-Based Chatbot

INDIVIDUAL THESIS

Kothalawala L S S

IT22185884

NLP Based Emotion Aware, Intent Aware Adaptive Chatbot

Group ID: R26-IT-018

B.Sc. (Hons) in Information Technology Specializing in

Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

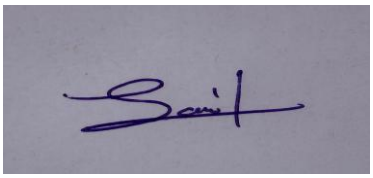
DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration. I declare that this thesis is my own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of my knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. I grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this thesis, in whole or in part, in print, electronic, or other media, while retaining the right to use the content in future work.

Academic integrity and AI disclosure. I have disclosed all use of generative AI and other automated tools in accordance with the module policy. I remain responsible for the accuracy, originality, citations, code, data, analyses, and conclusions in this thesis.

Student name: Kothalawala L S S

Student ID: IT22185884

Signature:  Date: ____2026/10/10_____

Supervisor certification. The above candidate carried out this research for the bachelor's degree thesis under my supervision.

Role and name	Signature	Date
Supervisor: Prof. Anuradha Jayakodi		
Co-supervisor: Mr. Dinith Primal		

ABSTRACT

Deaf and hard-of-hearing students studying Sri Lankan G.C.E. Ordinary Level Information and Communication Technology may require explanations that address their communication preferences, prior knowledge and learning difficulties. This research aims to explore the relationship between learner information and curriculum context to relevant chatbot explanations and revision support. The individual aim is to design, develop and evaluate an Emotion-Aware Adaptive Chatbot within an accessible e-learning platform.

The study adheres to a Design Science Research approach. The component utilizes student questions, selected feedback, conversation history, quiz performance and available webcam-derived indicators through an Emotion-Aware Response Adaptation engine. Learning resources are organized according to Grade 10 and Grade 11, each with three semesters. Responses feature simplified explanations, analogies, examination guidance and diagram- or image-based support when applicable. The implementation utilizes a React frontend, FastAPI backend and MongoDB, integrating a configurable language model with a local curriculum dataset fallback.

Developer testing included 12 functional cases, all of which passed, representing 40% of the planned 30-case functional evaluation. The completed checks addressed curriculum selection, response adaptation, revision activities, conversation retrieval, selected integration workflows and fallback operation. Controlled adaptation scores, response-time measurements and student and teacher feedback are not yet documented; therefore, educational effectiveness has not been established.

The individual contribution includes the adaptation workflow and its integration with curriculum resources, revision features and teacher-support information. The findings offer initial evidence of functional feasibility, while additional evaluation is required to determine the component's accessibility, response quality and usefulness to its intended learners.

Keywords: adaptive chatbot, deaf and hard-of-hearing learners, emotion-aware learning, accessible e-learning, O/L ICT

ACKNOWLEDGEMENTS

The author would like to extend his sincere gratitude to his supervisor and co-supervisor for their guidance, constructive feedback and encouragement throughout the development of his individual research component, the Emotion-Aware Adaptive Chatbot. The author appreciates their advice in formulating the research problem, methodology, system design and evaluation approach.

The author also thanks the Sri Lanka Institute of Information Technology (SLIIT) and the Faculty of Computing for the academic support and facilities provided during the research. The author is grateful to his research group members for their cooperation and assistance in the integration of the chatbot with the accessible O/L ICT e-learning platform. Their contributions and discussions enabled the author to address the connections between the individual components.

Finally, the author thanks his family and friends for their patience, understanding and encouragement throughout the research, development and thesis-writing process. Their support helped him remain committed to completing this work.

TABLE OF CONTENTS

1 INTRODUCTION 1

1.1 Overall project context 1

1.2 Individual component context 2

1.3 Problem statement 2

1.4 Research gap 3

1.5 Aim and objectives 3

1.6 Research questions or hypotheses 4

1.7 Scope limitations and assumptions 5

1.8 Individual contribution 5

1.9 Thesis structure 6

2 LITERATURE REVIEW AND RELATED WORK 7

2.1 Review method and scope 7

2.2 Domain and theoretical foundations 7

2.3 Related research 9

2.4 Existing systems products and techniques 10

2.5 Gap synthesis and conceptual framework 11

3 RESEARCH METHODOLOGY 13

3.1 Research design 13

3.2 Data participants or study materials 14

3.3 Experimental or study procedure 15

3.4 Measures and evaluation metrics 16

3.5 Data analysis 18

3.6 Ethics privacy security and responsible AI 19

3.7 Reproducibility and research artefacts 19

4 PRODUCT AND SOLUTION DESIGN 20
4.1 Requirements 20
4.2 Component architecture within the complete solution 22
4.3 Design alternatives and decisions 23
4.4 Detailed design 24
4.5 Standards and professional practice 29
5 IMPLEMENTATION INTEGRATION AND TESTING 31
5.1 Development environment and technology stack 31
5.2 Component implementation 32
5.3 Integration with the other components 33
5.4 Verification and software testing 33
5.5 Deployment and operation 35
6 RESULTS AND VALIDATION 36
6.1 Research results 36
6.2 Product and technical results 38
6.3 User or stakeholder validation 39
6.4 Integration results 39
6.5 Summary of findings 40
7 DISCUSSION 41
7.1 Interpretation of findings 41
7.2 Comparison with literature baselines and existing solutions 41
7.3 Research contribution 42
7.4 Product contribution 42
7.5 Limitations and threats to validity 43
7.6 Implications 43
8 COMMERCIALISATION AND SUSTAINABILITY 45

8.1 Target Users and Value Proposition 45
8.2 Commercialization Strategy 45
8.3 Sustainability and Maintenance 45
9 CONCLUSION AND FUTURE DIRECTIONS 47
9.1 Conclusion 47
9.2 Achievement of objectives 47
9.3 Individual research and product contributions 48
9.4 Future directions 48
REFERENCES 50
GLOSSARY 52
APPENDICES 53
Appendix A: Evaluation Instruments 53
Appendix B: Academic Integrity and AI-Use Disclosure 53

LIST OF FIGURES

Figure 2.1: Conceptual framework of the Emotion-Aware Adaptive Chatbot 12
Figure 3.1: Adaptive chatbot processing pipeline 13
Figure 4.1: Component architecture within the accessible e-learning platform 22
Figure 4.2: Adaptive response generation and local fallback workflow 27
Figure 5.1: Adaptive chatbot interface showing learning-state selection, topic context and revision support 32
Figure 6.1: Passed and failed functional test cases by evaluation area 37
Figure 7.1: Adaptive learning support and teacher follow-up loop 44

LIST OF TABLES

Table 2.1: Comparison of existing systems and techniques 10
Table 3.1: Planned evaluation materials and quantities 14
Table 3.2: Evaluation measures and assessment methods 17
Table 4.1: Functional requirements 20
Table 4.2: Non-functional requirements 21
Table 4.3: Design alternatives and selected approaches 23
Table 4.4: EARA input priorities and decision rules……..... 25
Table 4.5: Main chatbot API endpoints 28
Table 5.1: Implementation technologies and their purposes 31
Table 5.2: Integration with other platform components 33
Table 5.3: Functional test cases and observed outcomes 34
Table 6.1: Evaluation status at draft submission 36
Table 6.2: Summary of completed functional testing 36
Table 6.3: Functional evaluation metrics 37
Table 6.4: Product and technical outcomes 38
Table 6.5: Stakeholder validation focus 39
Table 9.1: Achievement of research objectives 47
Table A.1: Evaluation instruments and information collected 53

LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
DSR	Design Science Research
EARA	Emotion-Aware Response Adaptation
FR	Functional Requirement
G.C.E.	General Certificate of Education
ICT	Information and Communication Technology
JSON	JavaScript Object Notation
LLM	Large Language Model
NFR	Non-Functional Requirement
NLP	Natural Language Processing
O/L	Ordinary Level
OWASP	Open Worldwide Application Security Project
PDF	Portable Document Format
RAG	Retrieval-Augmented Generation
UI	User Interface
W3C	World Wide Web Consortium
WCAG	Web Content Accessibility Guidelines

1 INTRODUCTION

1.1 Overall project context

This research is dedicated to developing an accessible e-learning platform for deaf or hard-of-hearing students taking up G.C.E. Ordinary Level Information and Communication Technology (O/L ICT) in Sri Lanka. It seeks to develop learning assistance that can address their questions, focus, and difficulties.

Lessons in ICT contain concepts, terminology, and procedures that need to be explained clearly using multiple approaches. The learning materials were organized by grades 10 and 11 with each having three semesters. The developed e-learning tool can help students study the topics of their chosen grade and semester through text explanation, visual explanation, sign avatar lectures, and questions.

The project comprises four main contributions to the existing learning environment. These include attention tracker with knowledge graph questions, an emotion-aware chatbot with adaptive learning guidance, a sign avatar lecture generator, and a smart haptic wristband. Each component of the project accommodates the lessons' concepts, revisions, and non-audio communication in a shared space for a successful learning experience.

However, to present a comprehensive outlook of this project, this individual thesis will focus on the adaptive chatbot. It recognizes the syllabus category in a chosen grade and semester before providing an answer or recommendation. The chatbot uses students' questions, webcam information, selected feedback options, and quiz results for its explanation, learning guidance, and recommendation for Learning or Exam mode.

Depending on the context, it can recommend related concepts with simplified explanations, relatable examples, prerequisite skills, and practice problems. Furthermore, if the learning state suggests a lack of understanding or boredom, the chatbot can also share textual and visual (diagrams, images) explanations. The attention tracker uses questions or statements to raise awareness rather than outright confirming miscomprehension.

In short, this thesis discusses the design, implementation, and evaluation of the project. It aims to elaborate on the project's primary component, the chatbot, and its recommendation system. Specifically, it highlights how the explanation and revision recommendations are generated based on the curriculum and learning state information.

1.2 Individual component context

The Emotion-Aware Adaptive Chatbot provides interactive learning and revision within the module. It draws on information given by the students, the selected learning mode, preference of explanations and the other provided learner data to offer its responses. The learning material is divided into Grade 10 and Grade 11, each comprising three semesters.

The chatbot first identifies the relevant Syllabus category according to the chosen grade and semester before suggesting an explanation or revision plan. It evaluates the curriculum context with the student-selected feedback, webcam emotion indicators, chat text, and quiz results to determine the most relevant explanation and recommendation for the user in relation to their Learning or Exam mode.

This component uses an Emotion-Aware Response Adaptation (EARA) engine to alter its responses based on users' emotional expressions. When a student struggles to comprehend a concept, the chatbot offers to explain further using simplified concepts, prerequisite revision, or a diagrammatic explanation. Similarly, the chatbot can use visual and practical examples to engage the attention of students who demonstrate boredom through their webcam emotions. Meanwhile, students who show signs of comprehension can be offered more challenging questions to further test their understanding. Students can also choose to view standard, simplified, analogy-based, or examination-focused explanations.

The chatbot component uses backend Large Language Models (LLM) in combination with a local knowledge bank of O/L ICT curriculums, lessons, condensed notes, and question banks. The local backend system offers limited assistance when the chatbot is unable to access the designated LLM models. Recommendations are made based on attention values to ensure that the students effectively revise on the sections of the curriculum which they have struggled to grasp. This component also offers a feature for teachers to view micro-challenges, summaries, repeated questions, and progress tracking of the students in relation to the chatbot.

My contributions to this component include the logic for adaptive response generation, syllabus-integrated learning resources, visual response options, revision features, and teacher insights features. The component processes information from the wider platform while its primary function is focused on determining the most suitable method of offering learning support through the chatbot.

1.3 Problem statement

An ICT explanation may not resolve a student's difficulty if it fails to account for their understanding, communication needs or current syllabus topic. Deaf and hard-of-hearing learners may require simpler wording, diagrammatic or imagery-based explanations, or revision of prerequisite concepts. Repetition of the same text may not address a student's confusion or boredom.

The chatbot therefore needs to identify the relevant Grade 10 or Grade 11 syllabus category within the selected semester, and take the student's learning need into

account, before answering or recommending activities. It must exercise caution when using the available learning-state indicators, as attention patterns alone cannot confirm that a student understands a concept.

This research investigates the following problem: How can an adaptive chatbot combine student queries, learning-state indicators and curriculum context to provide appropriate explanations and revision support for deaf and hard-of-hearing O/L ICT students?

1.4 Research gap

The research gap that this study addresses focuses on learning-state information for guiding curriculum specific chatbot support for deaf and hard-of-hearing O/L ICT students. Detecting potential distraction or answering a question does not necessarily dictate what kind of explanation or revision activity would address the learner's difficulty.

In this study, we investigate how student questions, learning-state indicators, prior interactions and concept relationships can be integrated with the Grade 10 and Grade 11 syllabus content organized by semester. We examine both the selection and presentation of relevant content including diagrammatic or image-based explanations if the students are facing misunderstanding or boredom with the concepts.

This study also looks at how such a context can inform learning plans, Learning or Exam mode recommendations, follow-up practice and teacher intervention for recurring difficulties. This study makes a novel contribution by designing and evaluating these linked support strategies in the context of Sri Lankan O/L ICT learning environment.

1.5 Aim and objectives

Design an adaptive emotion-aware chatbot that uses the given query, learning-state indicators and curriculum context to provide accessible explanations and revision support for Deaf and hard-of-hearing students taking Sri Lankan G.C.E. O/L ICT.

- Organize syllabus-aligned lessons, short notes and question banks into Grade 10 and Grade 11, with three semesters per grade.
- Determine the syllabus category and the student's learning need before supplying answers, learning plan or Learning/Exam mode suggestions.
- Combine webcam analysis indicators, student feedback, chat history and quiz results to adapt the response.
- Develop EARA logic to adjust explanation style, length and difficulty, including diagram-based and image-based support for misunderstanding or boredom.
- Integrate LLM-based responses with a local dataset fallback to maintain assistance during model or service failures.

- Provide targeted revision through attention-based recommendations, prerequisite reminders, micro-challenges and spaced practice.
- Develop progress analytics and repeated-query alerts to support teacher intervention.
- Evaluate functionality, syllabus relevance, adaptation suitability, response time and usability through technical testing and student and teacher feedback.

Remark: Research objectives investigate how learner information and curriculum context can inform appropriate responses. System objectives address the practical application and integration of this support.

1.6 Research questions or hypotheses

The main research question guiding this study is:

How can an emotion aware adaptive chatbot use student queries, learning state indicators and curriculum context to offer suitable learning and revision support to deaf and hard of hearing O/L ICT students?

The following supporting questions examine the main aspects of the component:

- How can webcam-based indicators, student feedback, chat history and quiz results be incorporated to help response adaptation?
- How well does the chatbot identify relevant Grade 10 or Grade 11 syllabus category within the selected semester and employ the student's learning need to guide responses and recommendations?
- How effectively does the EARA adjust explanation styles, length and difficulty, including diagram- and image-based guidance for comprehension or boredom?
- How reliable is the local dataset fallback to provide relevant learning support if the specified LLM is unavailable?
- How useful are the chatbot's explanations, visual support, revision activities and progress information to students and teachers?

These questions form the basis of the controlled response evaluation, functional testing, response-time measurements and student and teacher feedback. The study considers response suitability, reliability and perceived usefulness. Claims concerning enhanced learning achievement would require additional evidence from a comparative learning assessment.

1.7 Scope limitations and assumptions

This study presents the Emotion-Aware Adaptive Chatbot including syllabus-aware explanations, response adaptation and revision activities, and teacher analytics. Resources will be arranged based on Grade 10 and Grade 11, each of which consists of three semesters. The chatbot will take the related syllabus category and learning need into account prior to giving answers or recommendations. It can provide diagram-based and image-based explanations for miscommunication or boredom, as well as large language model (LLM) responses and local dataset fallback.

The main limitations are:

- Curriculum coverage and default responses rely on the availability of learning resources and their accurate mapping.
- Learning indicators provide support requirements but neither prove nor disprove emotion or comprehension.
- Artificial text and images may contain inaccuracies which require validation.
- Mastery scores and mock examination grades represent support indicators rather than confirmed predictions of O/L grades.
- Tech test evaluation and user feedback neither confirm nor predict prolonged learning developments.
- Attention recognition, sign avatar creation and wristband design lie outside the scope of this component.

The study presumes the availability of appropriate learning resources, accurate matching of the learning resources to student records and the provision of adequate information from connected components. Webcam information requires appropriate conditions for recording. The chatbot is an addition to the learning experience, with teachers being responsible for addressing any alerts and providing additional learning support.

1.8 Individual contribution

My individual contribution is design, development and evaluation of Emotion-Aware Adaptive Chatbot. It is an application that links learner's information with O/L ICT curriculum content and gives appropriate explanations, revision activities and teacher support information.

My main responsibilities include:

- Organize syllabus material, lessons, short notes and question banks for chatbot use.
- Organize learning resources by grade, semester and syllabus category.
- Develop EARA logic based on student queries, learning-state indicators, interaction history and quiz performance.
- Implement explanation styles, Learning/Exam modes and diagram-based or image-based assistance for addressing misunderstanding and boredom.

- Ensure answers, learning plans and recommendations take into account the relevant syllabus category and learning need.
- Integrate backend LLM services with local dataset fallback.
- Develop revision activities, conversation storage, progress analytics and repeated-query alerts.
- Build the chatbot interface, integrate relevant platform services and evaluate functionality, response time and user feedback.

Other team members are working on the attention monitoring, knowledge graph, sign avatar and wristband components. My task is to incorporate their relevant outputs into the chatbot. The main research contribution is the design and evaluation of an adaptation process mapping learner context and syllabus content to appropriate learning support.

1.9 Thesis structure

This thesis is organized into nine chapters that provide information on the background of the research, the development and evaluation of the Emotion-Aware Adaptive Chatbot.

- **Chapter 1 - Introduction:** This section provides the project context, research problem, gap, aim, objectives, research questions, scope and individual contribution.
- **Chapter 2 - Literature Review and Related Work:** Literature Review and Related Work: Covers relevant existing theories, research and systems, representing the basis for the proposed solution.
- **Chapter 3 - Research Methodology:** Describes the research design, study materials, evaluation procedures, metrics and ethical considerations.
- **Chapter 4 - Product and Solution Design:** Explains the requirements, component architecture, design decisions and detailed chatbot workflow.
- **Chapter 5 - Implementation, Integration and Testing:** Details the development environment, implementation, integration with other components and software testing.
- **Chapter 6 - Results and Validation:** Reports the technical evaluation results and the findings of the student and teacher feedback.
- **Chapter 7 - Discussion:** Interprets the results and relates them to the research questions and related work, providing contributions and limitations of the results.
- **Chapter 8 - Commercialization and Sustainability:** Assesses the value of the platform, potential users, adoption, scalability and sustainability.
- **Chapter 9 - Conclusion and Future Directions:** Summarizes the study, discusses the fulfillment of the objectives and provides directions for future work.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This chapter reviews literature relevant to an emotion-aware adaptive chatbot for deaf and hard-of-hearing O/L ICT students. This review takes a thematic approach, covering a number of relevant areas such as accessible e-learning, educational chatbots, learning-state adaptation, curriculum-based support and personalized revision. Existing review articles on educational chatbots act as a springboard to explore the applications, benefits, and implementation challenges of such systems [1].

The articles are compared primarily based on the target learners, adaptation inputs, accessibility features, curriculum relevance and evidence of evaluation. Special attention is paid to the role of student questions, interaction history, assessment performance and emotional or attention clues, in guiding learning guidance. The use of texts, diagrams and images to explain concepts are also explored.

The review considers studies conducted with deaf and hard-of-hearing participants, as well as broader research conducted on adaptive tutoring. Findings from the latter are treated with caution, as their applicability to the target population of Sri Lankan O/L ICT students need to be explored further. In the context of this project, curriculum relevance entails identifying the appropriate topic (Grade 10 or Grade 11) within the semester structure of the platform prior to providing explanations or recommendations.

Finally, the review highlights the difference between implemented features and measured outcomes, providing a foundation to explore the synergy between syllabus context, learner information and visual explanations within the confines of the chatbot. This review is not intended to be an exhaustive or systematic review of the research literature in this area but rather a focused thematic review.

2.2 Domain and theoretical foundations

The Emotion-Aware Adaptive Chatbot makes use of multimedia learning, emotion-aware tutoring, curriculum-based support and retrieval practice. These will inform how explanations are delivered by the chatbot and how revision activities relate to a student's needs.

Accessible learning and multimedia presentation

The Cognitive Theory of Multimedia Learning by Mayer lists three assumptions – separate channels for processing, a limit to processing capacity and active processing of information [2]. These underpin the idea of presenting explanation in digestible sections and linking language to relevant visuals. However, the assumptions about auditory and visual processing need to be re-examined when designing for deaf students.

In light of this project, the implication will be to combine clear text with relevant diagrams or images while limiting superfluous visual information. For instance, an explanation of the input–process–output cycle can be supported by a labelled diagram and a brief description of each step. Students should be able to navigate the explanation at their own pace. Reacting to visual stimuli in response to confusion or boredom is a design decision that must be explored, rather than an assumption that images enhance learning.

Emotion-aware adaptation

Emotion-aware tutoring regards how a learner's emotional state can inform teaching and support. D'Mello *et al.* observe that confusion can be a useful part of the learning process, so long as it is induced, regulated and resolved [3]. Their experiments involved scientific reasoning with animated conversational agents; there is no assumption that confusion is universally beneficial or applicable to deaf O/L ICT students.

This nuance is acknowledged in the EARA approach. A student may need a simpler explanation, a reminder of prerequisite information or an alternate perspective on the issue. The chatbot will take into consideration webcam-derived indicators as well as student feedback, chat logs and performance on quizzes. Attention, emotion and understanding are viewed as separate, simultaneous elements.

Curriculum context and prerequisite support

The current project organizes learning resources by Grade 10 and Grade 11, with three semesters each. In context of relevant syllabus category and student need, explanations, learning plans and Learning/Exam mode recommendations will be generated.

Concept relationships will support prerequisite revision. For example, a student who finds a logic-gate truth table challenging may need to review binary input and output values. A short reminder or micro-challenge can confirm the student's prior knowledge before the chatbot proceeds to the main topic. Grade and semester will establish the curriculum context – they do not determine the student's level of mastery.

Retrieval practice and revision

Retrieval practice refers to the act of recalling information, such as answering questions or performing tasks. Karpicke and Roediger found that repeated retrieval practice improved retention of foreign language vocabulary in an experiment [4]. This finding supports the inclusion of micro-challenges, diagnostic quizzes and flashcards to confirm understanding, identify areas for improvement and promote further practice. The effectiveness of retrieval practice for deaf and hard-of-hearing O/L ICT students is a matter for further consideration in the design project.

2.3 Related research

Research on educational chatbots, affective tutoring and knowledge-supported generation informs the design of this component. The following studies provide valuable approaches while demonstrating the need for targeted evaluation within the intended learning context.

Educational chatbots and conversational tutoring

Okonkwo and Ade-Ibijola reviewed educational chatbot applications identifying their uses, benefits and implementation challenges [1]. While their review offers a broad foundation for exploring conversational learning support, it does not measure the effectiveness of this project's particular combination of accessibility features and syllabus-aware adaptation.

D'Mello and Graesser's AutoTutor and Affective AutoTutor work investigates the learning that takes place through dialogue with cognitively and emotionally responsive computer tutors [5]. This provides a relevant precedent for EARA's use of learner context to inform its responses. The current component applies a similar approach to Grade 10 and Grade 11 ICT topics with an emphasis on selectable explanation styles and visual support. These adaptations require independent evaluation with deaf and hard-of-hearing students.

Confusion and adaptive support

D'Mello *et al*. investigated confusion occurring during scientific-reasoning dialogues finding it beneficial to learning when appropriately induced, regulated and resolved [3]. This suggests a tutor should analyze the nature of a difficulty before determining an appropriate response.

For this chatbot, possible responses might include prerequisite reminders, micro-challenges or an alternative visual explanation. However, no study validates the rule of providing diagrams or images during misunderstanding or boredom established as a design decision for this project.

Retrieval-supported response generation

Lewis *et al*. introduced Retrieval-Augmented Generation (RAG) which combines a language model with an external document index. Their models achieved strong results on open-domain question-answering tasks and produced more factual language than a compared model using only its stored parameters [6].

This research suggests investigating how referring to relevant source materials could inform answers. It does not establish that generated answers will align with any particular syllabus. In this project, grade, semester and syllabus category provide that context. RAG also differs from local fallback: RAG uses retrieved content during generation while the fallback supplies an alternative answer when the configured LLM is unavailable.

Retrieval practice

Karpicke and Roediger demonstrated improved delayed recall through repeated retrieval in a vocabulary-learning experiment [4]. This supports the inclusion of active recall through quizzes, micro-challenges and flashcards. It does not directly validate the current chatbot's review intervals, mastery scores or overall effectiveness for O/L ICT learners.

These studies together inform this component's design while demonstrating the need to evaluate its effectiveness within the intended learning context. The project measures syllabus relevance, adaptation suitability, reliability and perceived usefulness as part of this assessment.

2.4 Existing systems products and techniques

Existing systems provide valuable methods for organization of course material, sign-language access and adaptive assistance. Table 2.1 shows the comparison between selected examples and the requirements for this chatbot. The comparison is based on documented capabilities rather than a performance benchmark.

Table 2.1: Comparison of existing systems and techniques

System or technique	Relevant capability	Implication for this project
Moodle	Controls access to activities and course sections based on conditions such as completion, grades and group membership [7].	Demonstrates structured learning progression. The chatbot also takes learner context into account when presenting explanations within the selected grade, semester and topic.
Hand Talk	Provides avatar-based website translation, with documented support for American Sign Language and Brazilian Sign Language [8].	Provides an example of sign-language assistance alongside website content. Its documented language support does not confirm suitability for Sri Lankan learners.
Affective AutoTutor	Uses a conversational tutor that responds to cognitive and emotional aspects of learning [5].	Informs EARA's use of learner context. The project applies its own adaptation rules to O/L ICT explanations, visual

		support and revision.
Retrieval-Augmented Generation	Combines language generation with information retrieved from an external knowledge source [6].	Informs the use of relevant source material. It differs from local fallback, which provides an alternative answer when LLM generation is unavailable.

Moodle's restrictions to the access of learning resources by students are defined; this feature is different from the ability to edit an explanation within a conversation. Hand Talk application demonstrates how sign language translation is integrated into media, but the accuracy of local language interpretation and its pedagogical value requires additional consideration.

Affective tutoring and retrieval-based generation cover distinct aspects of the learning-support system: the former is aimed at the response transformation, while the latter includes the use of external information for refining answers. The current component combines these two directions in the processes of syllabus identification, visual explanation, revision activities and teacher analytics. Their inclusion is the project's evaluation focus; the comparison does not indicate the chatbot's superiority over the mentioned solutions.

2.5 Gap synthesis and conceptual framework

The reviewed literature describes emotion-aware dialogue [5], retrieval-supported generation [6] and retrieval practice [4], and existing products describe conditional learning pathways [7] and sign-language assistance [8]. But the reviewed evidence does not show how such approaches support deaf and hard-of-hearing students studying Sri Lankan O/L ICT.

The gap relates to connecting the learner information with an appropriate instructional response. An attention indicator does not specify the concept for revision, nor does a relevant answer specify that the student understands it; providing a diagram or image does not ensure that it is addressing their difficulty.

This study combines learner information with the relevant grade, semester, syllabus category and prerequisite relationships. The conceptual framework contains four stages:

- **Learner inputs:** Student questions, selected feedback, webcam-derived indications, conversation flows and quiz results inform us about their situation.

- **Context and adaptation:** Identify relevant Grade 10 or Grade 11 syllabus category and learning need, EARA uses this context to determine explanation style, depth, difficulty, and mode recommendations
- **Learning support:** The chatbot provides explanations, learning plans and revision activities using the configured LLM or local fallback, diagram or image-based explanations provide additional support for misunderstanding or boredom.
- **Feedback and follow-up:** Student responses and activities inform subsequent assistance, progress tracking and repeated-query alerts for teachers

Figure 2.1 illustrates these relationships. The LLM and local fallback are alternative answer sources within the learning-support stage.

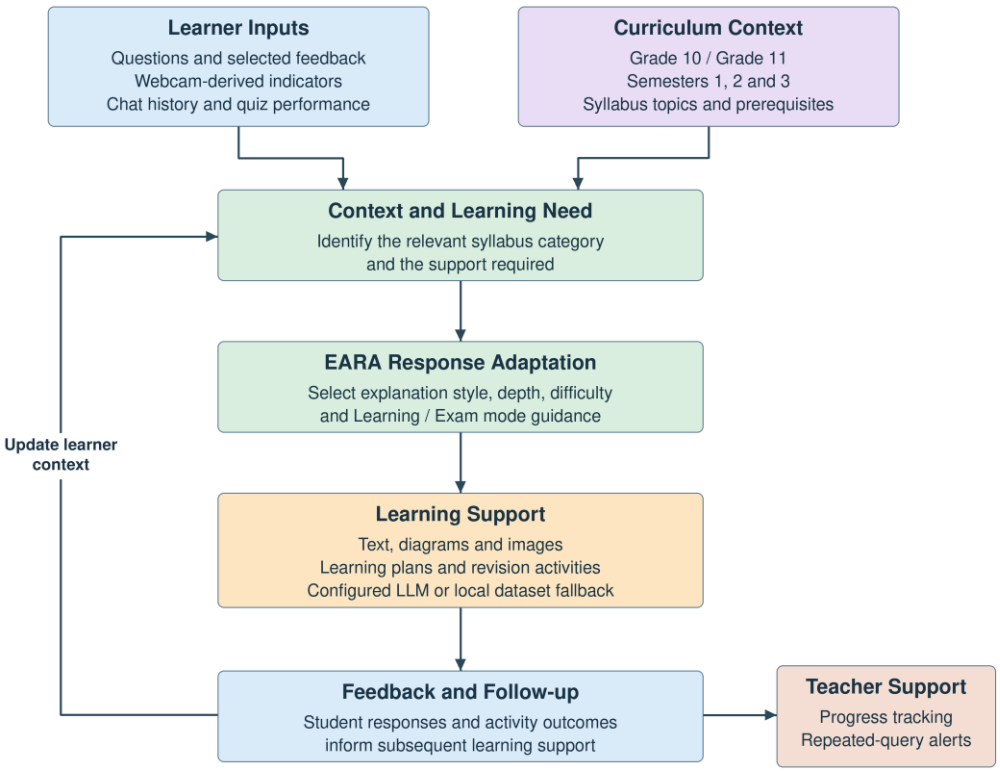


Figure 2.1: Conceptual framework of the Emotion-Aware Adaptive Chatbot

The framework treats learning-state indicators as evidence for selecting assistance rather than definitive measures of emotion or understanding. Its suitability is examined through syllabus relevance, response adaptation, technical performance and student and teacher feedback.

3 RESEARCH METHODOLOGY

3.1 Research design

This study follows a Design Science Research (DSR) methodology to elaborate the design of Emotion-Aware Adaptive Chatbot. The approach is described in terms of problem identification, solution objectives, design and development, demonstration, evaluation and communication activities [9]. These activities are applied to establish the links between the identified learning-support needs and the process of building and iterating the chatbot design.

The chatbot processing pipeline starts by taking the initial question of the student and information related to the learner such as webcam-inferred indications, chosen feedback, conversation history and quiz responses as the input. In order to offer help, the system needs to identify the relevant Grade 10 or Grade 11 syllabus category in the chosen semester and the learning need of the student. The lesson content and the prerequisite associations offer further context.

The Emotion-Aware Response Adaptation (EARA) engine then guides explanation style, detail and difficulty. For misunderstanding or boredom, support may include diagram-based or image-based explanations with accompanying text. The configured LLM provides the primary response path, while a local dataset fallback maintains support during service failures.

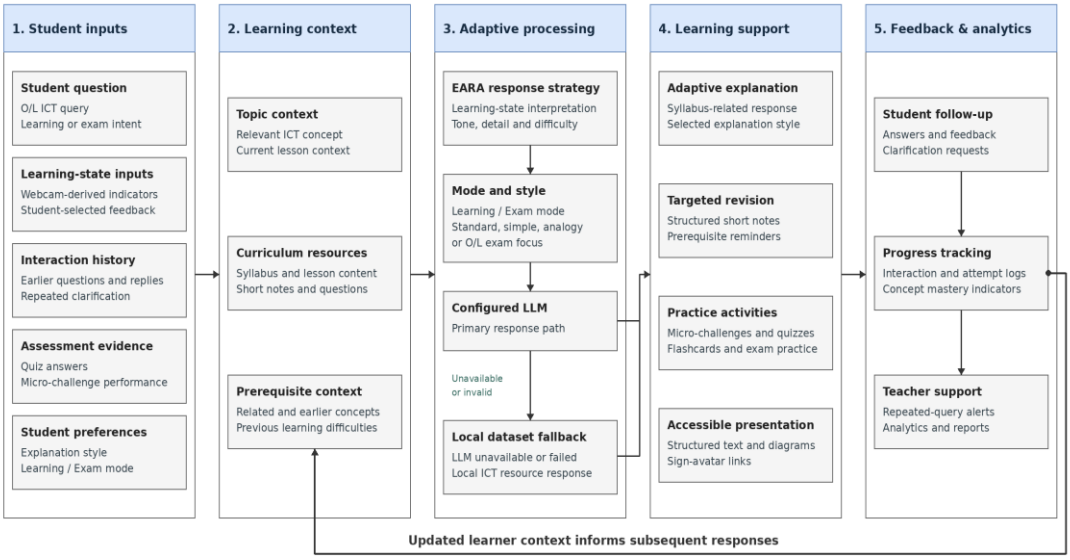


Figure 3.1: Adaptive chatbot processing pipeline

The pipeline connects explanations, learning plans and revision activities with subsequent student interactions that provide the basis for future assistance, progress

tracking and repeated-query alerts. Figure 3.1 represents the operational workflow; the DSR approach provides the overall method for developing and evaluating it.

The technical evaluation proposed involves 96 controlled scenarios, calculated as six grade–semester groups × two question contexts × four learning states × two interaction modes. This provides 48 scenarios each for grade (50%) and learning state (25%). The 48 scenarios involving misunderstanding and boredom therefore allow visual-response behavior to be investigated where applicable. Note that these percentages describe test coverage, not measured effectiveness.

A further 30 functional test cases will investigate features and integration behavior. 60 timed requests will be divided equally between successful LLM responses, direct local responses and failure-triggered fallback (20 requests each). Student testing and teacher feedback will investigate clarity, syllabus relevance, visual usefulness and ease of use. Participant details will be reported from actual evaluation records.

Controlled scenarios evaluate response adaptation under specified conditions, but do not constitute independent validation of learning-state detection or demonstration of improved learning achievement.

3.2 Data participants or study materials

The study uses O/L ICT curriculum resources, learner interaction information and evaluation materials. The curriculum contains syllabus data, lesson documents, short notes and question banks organized by Grade 10 and Grade 11 (each with three semesters). The resources were implemented to support explanations, lessons plans, revision activities and local fallback answers. Diagram- and image-based learner responses are included in the materials related to visual learning support.

The information collected from learners includes their questions, selected feedback, chat history, quiz performance and webcam-derived indicators provided by the monitoring component. The data is used by EARA to adapt support in line with four possible states: not understanding, being distracted, boredom and understanding. The states are not necessarily linked to emotions but rather explain the progress in learning that can be supported by the system.

The evaluation contained 96 controlled scenarios: two question contexts from each of the six grade-semester groups, implemented across the four learning states and in two modes of interaction. In turn, each state occupies 24 scenarios (25%) while Learning and Exam modes – 48 scenarios each (50%).

Table 3.1: Planned evaluation materials and quantities

Evaluation material	Planned quantity	Purpose
Question contexts	12: two per grade–semester group	Provide balanced curriculum coverage

Controlled chatbot scenarios	96	Assess syllabus relevance and adaptation across states and modes, including visual support
Functional test cases	30	Check features, integration and failure handling
Timed requests	60	Measure response time across three response paths

The timed requests are equally divided between successful LLM responses, direct local responses and failure-triggered fallback: 20 requests per condition (33.3%). The quantities refer to test plan descriptors rather than completed results.

Student testing will examine explanation clarity, visual usefulness, navigation and revision support. Teacher feedback will address subject accuracy, syllabus relevance and instructional suitability. Participant counts, recruitment details and learner characteristics will be reported from the actual evaluation records, distinguishing between deaf and hard-of-hearing O/L ICT students and any other participants.

3.3 Experimental or study procedure

The proposed evaluation involves four phases: preparation, controlled response testing, functional and response-time testing, and student and teacher evaluation. In this case, twelve reported functional cases, whereas the others are the description of planned procedures, which have been presented as the experiment steps.

Preparation

The research will apply twelve O/L ICT question contexts – two per grade-semester groups. The expected answer points, syllabus categories, and adaptation criteria will be identified during preparation before conducting the controlled tests. In addition, the application version, large language model (LLM) provider and model, timeout, and test environment will also be organized to ensure repeatable testing.

Controlled response testing

Each question context will be tested for every learning state and interaction mode, which will provide 96 scenarios. In this stage, grade, semester, topic, conversation history, and explanation-style settings will remain the same for comparison between the learning states. Each scenario within one question context will be run in a separate session, which will apply the same predefined conversation history to avoid interference from previous sessions.

The responses will be evaluated for subject accuracy, curriculum relevance, and adaptation appropriateness. Besides, if the diagrams/images are used to illustrate the explanation or prevent confusion or boredom, their relevance and consistency with the text will be assessed. Notably, controlled state labels have been applied to test EARA's response behavior rather than webcam-based learning state detection accuracy.

Functional and response-time testing

The functional test plan includes 30 cases to follow the syllabus, adaptive response, diagram/image, revision activities, conversation storage, analytics, integration capabilities, and fallback scenarios. Twelve reported functional test cases have been executed and passed, as described in section 5.4. However, the results of the 18 remaining cases were undocumented.

The response-time testing will involve 60 requests, including 20 successful LLM responses, 20 direct local responses, and 20 failure-triggered fallback responses. The time taken for each response to appear on the screen will be captured and analyzed. The measurement will start when a request is initiated and stop when the entire response, including the visual content, has been presented. The fallback responses' time will include the duration of failed LLM responses and extra timeout or retry delays, if any. The response format will also be captured to help explain the observed variations.

Student and teacher evaluation

The students will be able to pick a grade, semester, and topic, ask for clarification, change the explanation style, and complete a revision activity. The evaluation will be done based on explanation clarity, diagrams/images, navigation, and curriculum relevance. The teachers will review the selected responses to check accuracy and appropriateness. The participants' attributes, task results, ratings, and comments will be captured and analyzed together with the technical results to identify insights. The student and teacher evaluations have not been documented yet; therefore, no conclusions about stakeholder acceptance or improvements to learning can be drawn.

3.4 Measures and evaluation metrics

The evaluation considers response quality, curriculum relevance, adaptive behavior, reliability and usability. Table 3.2 defines the measures used for completed functional testing and the wider planned evaluation.

Table 3.2: Evaluation measures and assessment methods

Measure	Evaluation method
Answer quality	Teachers will score the accuracy, relevance and clarity of subjects separately from 1 (poor) to 5 (excellent).
Syllabus alignment	Responses will be evaluated against the expected grade, semester and syllabus topic, and the percentage of the defined criteria will be reported.
Adaptation suitability	Responses will be scored on a set of predefined criteria for each learning state and interaction mode, and the percentage of these criteria that will be achieved will be calculated.
Visual response quality	Where visuals are provided, teachers will rate accuracy, readability, relevance and consistency with the accompanying text on a five-point scale.
Functional correctness	Executed cases will be classified as passed or failed according to their expected outcomes.
Response time	Median and 95th percentile times will be reported separately for successful LLM responses, direct local responses and failure-triggered fallbacks. Timing includes the display of any accompanying visual content.
Fallback reliability	The percentage of simulated LLM failures that returned a valid local answer will be measured. Content relevance will be evaluated separately.
Usability and perceived usefulness	Student task completion, five-point feedback ratings and comments will assess the navigation, clarity of explanations and support in revision.

The functional pass rate will be calculated as:

$$\text{Functional pass rate} = \frac{\text{Number of passed cases}}{\text{Number of executed cases}} \times 100$$

Completion of the planned functional evaluation is calculated separately:

$$\text{Functional test completion} = \frac{\text{Number of executed cases}}{\text{Number of planned cases}} \times 100$$

Adaptation criteria will be based on the intended response types; for instance, a not-understanding response would be adapted by providing a simpler hint or prerequisite help with the task, possibly with a relevant visual aid. In turn, an understanding response will be adapted by providing a sufficiently detailed explanation or an alternative task. However, adaptation through mere wording changes will be deemed ineffective.

The functional pass rate and test completion rate were based on the twelve reported cases, as was the completion rate for the test. Other metrics are related to the general evaluation, which does not yet have specific measurement data. The rating will be based on valid responses, while the visual assessment will consider only responses that include the visual aid. The two quantitative measures are indicators of the system's performance and user evaluation but cannot independently demonstrate the improvement of learning outcomes.

3.5 Data analysis

The analysis highlights functional testing completed versus planned in broader terms. The 12 reported functional outcomes were summarized using passed and failed counts, the pass rate of executed cases and the proportion of completed 30-case planned test set. Cases were grouped by functionalities to identify which workflows were covered. These results reflect behaviors under specific tested conditions rather than an overall system effectiveness.

For the planned 96 controlled scenarios, the test logs will be scanned for missing entries, accidental duplicates and incomplete responses. Curriculum-alignment and adaptation success rates will be summarized by grade-semester group, learning state and interaction mode. Responses to the same question will be examined at fixed other settings. Teacher ratings will be presented using distributions and medians. Responses will also be reviewed for visual accuracy, readability, relevance and textual consistency.

Response times will be analyzed separately for successful LLM responses, direct local responses and failure-triggered fallback responses. Median and 95-th percentile values will describe typical and slower responses. As 20 requests will be used per condition, the statistics will be treated as performance estimates. Failed requests will continue to be included in the reliability analysis, while the number of successful time-bound responses will be reported for each condition.

Student and teacher feedback will be summarized using counts, percentages and median ratings. Written comments will be grouped into recurring themes, such as explanation clarity, navigation difficulties, curriculum relevance and revision usefulness. Missing feedback will be reported, with percentages calculated from valid responses to each item.

Only the functional outcomes are currently available for analysis. No statistical significance, stakeholder acceptance or learning-improvement claims are made from these results.

3.6 Ethics privacy security and responsible AI

- Learning-state indicators will be used as approximations rather than definitive statements about the emotions, understanding or ability of a student.
- Participation will be voluntary, with readily available consent information and the option to opt out. Parental/guardian consent and student assent will be requested for participants under 18.
- Research records will contain participant codes rather than names and will be accessible only to authorized researchers. Raw webcam recordings will not be retained for the purposes of the chatbot evaluation, or transmitted to external LLM services.
- API keys will be kept on the backend. External requests will not include unnecessary personal information, and participants will be informed about external processing.
- Students will be able to toggle between explanation styles and report inappropriate responses. Diagrams and images will be reviewed for accuracy, appropriate labelling and consistency with the explanation.
- Chatbot answers, automated marks, learning plans and revision recommendations will be used to support, but not replace, teacher guidance.

3.7 Reproducibility and research artefacts

The research artefacts will consist of the syllabus dataset, grade-semester topic mappings, lesson resources, adaptation rules, prompts templates, API specifications and evaluation records. The application and dependencies versions, LLM provider and model, timeout settings and test environment will be recorded for reproducibility purposes.

The 96 controlled scenarios, 30 functional test cases and 60 timed requests records will contain the input, expected and actual results. The relevant diagrams and images will be retained with their corresponding answers to enable assessing the visual quality. The student and teacher feedback will be anonymized prior to being included in the supporting materials.

The LLM requests repetition may result in varied outputs in terms of wording and visuals. The reproducibility, therefore, will be focused on the consistent test conditions, syllabus alignment, and adaptation behaviors rather than identical results. The shared artefacts will not include API keys, personal data and learning materials that cannot be redistributed.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The Requirements address syllabus-aware learning support, response adaptation, accessibility and teacher monitoring. Table 4.1 and 4.2 give the functional and non-functional requirements that will be used to guide design and test the system.

Functional requirements

Table 4.1: Functional requirements

ID	Requirement
FR01	Allow students to ask O/L ICT questions and get answers based on available syllabus and lesson material.
FR02	Use Webcam based indicators, student selected states, chat context and results of quizzes in response adaptation.
FR03	Support Learning and Exam modes, with normal, simplified, analogy based and exam focused explanations.
FR04	Provide local dataset answer if configured LLM not available or returns invalid response.
FR05	Provide lesson summaries, short notes, micro challenges and revision activities related to selected topic.
FR06	Recommend concepts for revision based on attention records and past learning activity.
FR07	Maintain conversation history, provide topic mastery indicators and learning progress.
FR08	Provide teacher analytics and repeated query alerts to identify students in need of additional support.
FR09	Support visual explanations and links to available sign avatar previews for relevant ICT terms.
FR10	Organize learning resources into Grade 10 and Grade 11, with three semesters each, and allow grade, semester and topic selection.
FR11	Identify relevant syllabus category and student learning need before giving answers, learning plans or Learning/Exam mode recommendations.
FR12	Provide relevant diagram based or image based explanations with supporting text when responding to misunderstanding or boredom.

Non-functional requirements

Table 4.2: Non-functional requirements

ID	Requirement
NFR01 - Accessibility	Present essential instructions and feedback via visual means using easily digestible text, intuitive navigation and accessible controls that are not audio-dependent.
NFR02 – Reliability	Gracefully handle LLM errors and explicitly identify when answers are generated from the local dataset.
NFR03 - Performance	Ensure responsiveness of interactions through configurable timeouts and measure response times separately for LLM and fallback pathways, including visual rendering.
NFR04 - Privacy and security	Secure student records, enforce role-based access controls and keep API credentials on the backend.
NFR05 – Maintainability	Separate interface, adaptation, data-ingestion and analytics logic so that each can be updated independently.
NFR06 - Integration	Exchange learner context and curriculum identifiers with other platform components via predefined API contracts.
NFR07 — Visual clarity	Display diagrams and images with appropriately labeled elements, clear relationships and associated text that consistently conveys the same information.

These requirements will guide the components design and provide the basis for functional testing and user evaluation.

4.2 Component architecture within the complete solution

The technical design of the Emotion-Aware Adaptive Chatbot application implies a multi-tier system with a React frontend, FastAPI backend and data layer. The internal structure of the system and its integration points with the broader learning platform are visualized in the block diagram of figure 4.1.

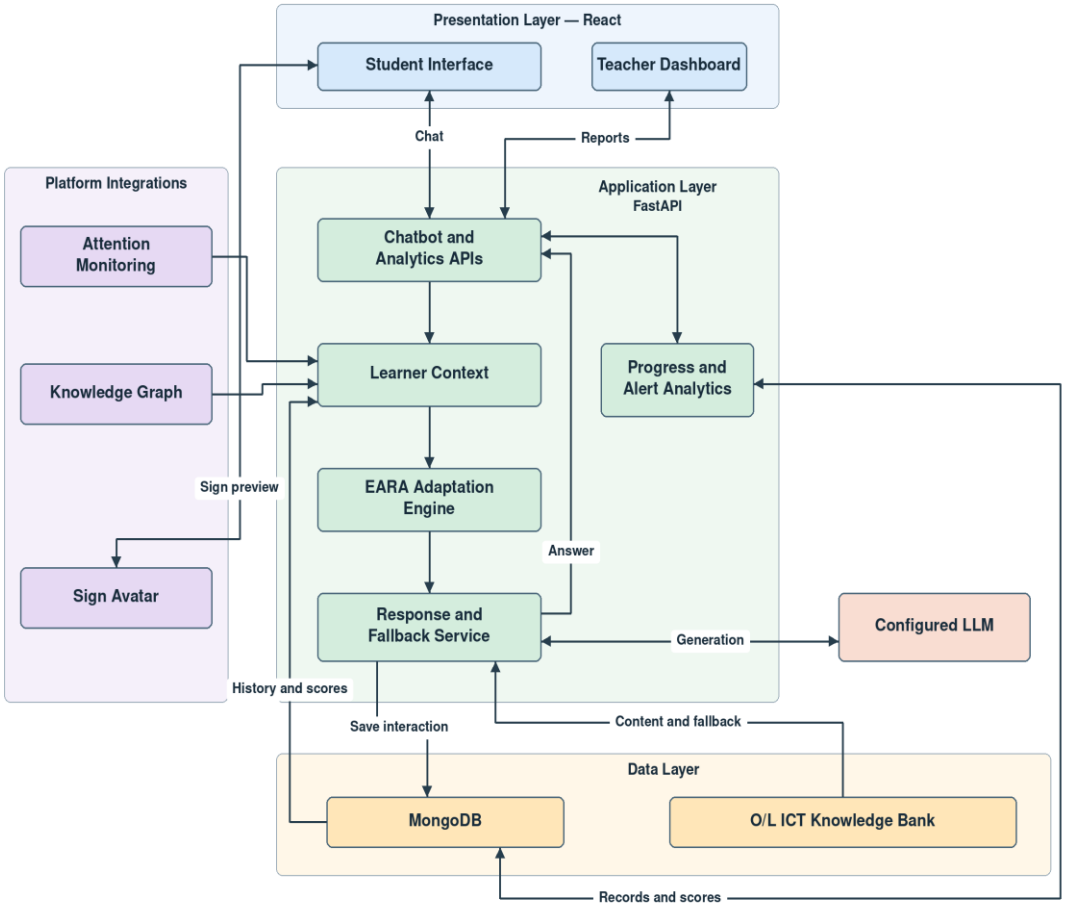


Figure 4.1: Component architecture within the accessible e-learning platform

Presentation layer: The student interface offers grade, semester, and topic selection, chat, explanation-type controls, revision activities, and progress views. Answers can include text, diagrams, and images. The teacher dashboard displays learning analytics and repeated-query warnings. Both interfaces communicate with the backend via API request.

Application layer: The chatbot service combines the student's question, selected feedback, conversation history, quiz results, and attention indicators. It determines the relevant syllabus category according to the selected grade and semester and analyzes the student's learning needs to prepare an answer or recommendation. EARA advises on the tone, level of detail, and further assistance, including visual

explanations for the observed lack of understanding or boredom. The response service uses the configured LLM, which has access to the local dataset in case of failure. Information about the sources of the response is provided alongside the answer.

Data layer: The MongoDB database stores the chatbot's conversations, messages, understanding scores, and repeated-query warnings. The O/L ICT knowledge bank includes the syllabus, lessons, short notes, and question banks organized by Grade 10 and Grade 11, each covering three semesters. The stored information helps the chatbot provide curriculum-related explanations, learning plans, and revision activities.

Integration with the complete solution: Attention-monitoring outputs support the identification of the segments of the lesson that the student should revise. In addition, the knowledge graph's relationship data support the provision of explanations related to the topic or prerequisite lessons. ICT keywords that the chatbot understands are linked to preview sign avatars. Furthermore, the records of the chatbot's conversations and the outcomes of the quizzes influence the future responses of the chatbot and the teacher's monitoring.

4.3 Design alternatives and decisions

The chatbot design features flexible explanations, syllabus relevance, accessibility and reliable operation. Table 4.3 shows the alternatives and the rationale behind the selected approaches.

Table 4.3: Design alternatives and selected approaches

Design area	Alternatives	Selected approach and justification
Response generation	Fixed answers, LLM only responses, or LLM with local fallback	A configurable LLM provides multiple explanations, and local fallback provides basic assistance when it fails. Coverage of fallback is dataset-dependent.
Response adaptation	A common explanation style or context-based adaptation	EARA adjusts the tone, detail level, difficulty and follow-up support based on learner context. Explicit rules make the intended behavior easier to test and review.
Learning-state inputs	Webcam indicators alone or multiple sources	Webcam derived indicators are combined with student-selected feedback, chat history and quiz results to reduce reliance on any one indicator.

Student control	Fully automatic adaptation or adaptation with manual controls	Students choose an explanation style and interaction mode that best fits their requirements, letting them request the support they need.
Learning content	General-purpose answers or syllabus focused resources	O/L ICT syllabus information, uploaded lessons, notes and question banks form the subject context for explanations and revision.
Curriculum organization	A single resource collection or grade semester topic organization	Resources are organized into Grade 10 and Grade 11, each covering three semesters. Syllabus category and learning need determines answers, learning plans and mode recommendations.
Explanation format	Text only responses or text with visual support	Diagrams and images supplement text explanations for clarification or to reduce tedium. Applicability, clarity and accuracy need to be assessed.
Service integration	Direct frontend LLM calls or backend managed calls	Backend managed integrations protect API credentials and manage timeout handling, response validity checks and fallback behavior.

The design decouples curriculum and learner-context preparation, response adaptation and answer generation. This enables independent updates to learning resources, adaptation rules and LLM configuration. It also facilitates testing whether an unsuitable answer is due to incorrect topic selection, inappropriate adaptation or response-generation failure.

4.4 Detailed design

A detailed design, which ties together curriculum selection, learner context, response adaptation and revision support, defines the stages by which the content of an answer and the presentation of that answer to the student is decided.

4.4.1 Learner Context and Response Adaptation

The chatbot uses the selected grade, semester and topic to define the curriculum context. EARA than takes into account the student's question, selected feedback, recent quiz results, conversation history and available webcam-based indicators to decide which support is appropriate.

The decision policy prioritizes explicit requests and feedback from the student. Indicators derived from indirect sources provide additional context but should not be used to form definite conclusions about the student's understanding. Table 4.4 defines the input priorities and rules for resolving conflicting information as specified for the component.

Table 4.4: EARA input priorities and decision rules

Input	**Priority and decision rule.**
Current question and explicit request	Determine the immediate task. Requests such as "explain simply" or "show a diagram" guide presentation for that response.
Student-selected learning state	Provides the primary stated support need, unless the current message explicitly requests something different.
Quiz performance	Supplies concept-specific evidence. An incorrect answer introduces corrective feedback or prerequisite support without automatically changing the student's selected state.
Conversation history	Helps identify unresolved questions, avoid repeating an unsuccessful explanation and maintain topic continuity.
Webcam-derived indicators	Provide supplementary cues. They do not override explicit feedback or independently establish misunderstanding.
Selected mode and explanation style	Guide the response format. A direct request for a different format applies to the current answer without changing the saved preference.

Conflicting and missing information

If a student selects understanding but incorrectly answers a quiz, the specified response provides feedback on the incorrect concept and includes an extension question. The selected state is retained. If the webcam indicators suggest a student is distracted, but the student indicates they understand, EARA retains the indicated learning state and offers a brief recap or optional check.

If an input is unavailable, it should be omitted from consideration instead of assuming a neutral or positive state. If webcam observations are unavailable, the adaptation could continue using the question, chosen feedback and learning records. With insufficient learner information, the chatbot offers a standard curriculum-based explanation and asks if a simpler or visual explanation would help. If the curriculum context is unclear, it asks for clarification before recommending topics.

Response selection

For not understanding, a response may use simpler wording, remind of prerequisites and include relevant visual examples. For bored, a relatable context is introduced with a visual example or challenge. For distracted, a brief instruction focused on one task is offered. For understanding, more depth or an application question can be offered.

The specified policy uses explicit rules rather than a numerical fusion score. It does not introduce numerical confidence or quiz-percentage thresholds. This behavior needs to be checked and tested in implementations to confirm this behavior.

4.4.2 Answer Generation and Fallback

The response service uses the prepared curriculum and learner context to request an answer from the configured LLM. Backend processing handles credentials, timeout settings and response validation.

When no LLM is configured, the local dataset contains the response path. It also acts as a fallback after timeout, authentication, quota, network, server or invalid-response failures. Local answers are dependent on the coverage of the available syllabus resources.

Figure 4.2 depicts this process. A technically usable LLM response progresses to answer preparation, while an unsuccessful request is redirected to the local dataset. Technical validation does not confirm factual accuracy; content quality is evaluated separately during evaluation.

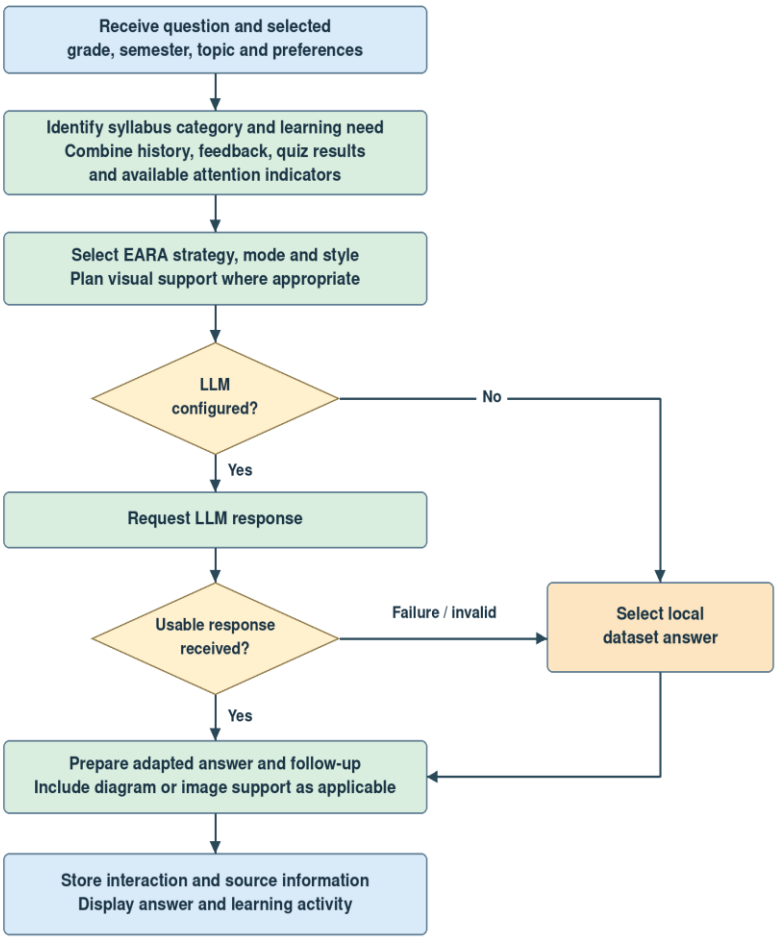


Figure 4.2: Adaptive response generation and local fallback workflow

The response contains source type, fallback reason and confidence metadata. These fields describe the response source and processing outcome. The confidence value is interpreted as a system metric rather than a validated probability that the answer is correct.

4.4.3 Data and Revision Support

MongoDB stores chatbot sessions and messages in chatbotSessions and chatbotMessages. Associated collections store data related to syllabus topics, summaries of lessons, micro-challenges, understanding scores and repeated query alerts. Resources related to curriculum are tied to their grade, semester and topic to enable relevant revision.

Attention-based recommendations relate missed parts of a lesson to relevant concepts. Notes, prerequisites, flashcards and practice questions act as follow-up activities. Subsequent response and actions contribute to progress tracking and the context for future answers.

Repeated issues demonstrate understanding problems with a particular concept, triggering teacher alerts. Teacher alerts and progress summaries highlight areas of concern, leaving the interpretation of student needs to the instructor.

4.4.4 API Interface Design

The chatbot provides API end points to support adaptive answers, resource selection for revision, revision activities and teacher analytics. Requests contain parameters required for the operation, and responses return relevant content, learning records or evaluation. Table 4.5 provides a summary of primary interfaces.

Table 4.5: Main chatbot API endpoints

Method	Endpoint	Purpose
POST	/api/chatbot/chat	Submit a question with learner and curriculum context and receive an adaptive response.
GET	/api/chatbot/topics	Retrieve supported ICT topics for curriculum selection.
GET	/api/chatbot/attention-recommendations/{student_id}	Retrieve revision recommendations based on attention records.
GET	/api/chatbot/short-notes/{topic_id}	Retrieve structured revision notes for a topic.
POST	/api/chatbot/challenge/check	Check a micro-challenge answer.
GET	/api/chatbot/knowledge-growth/{student_id}	Retrieve topic mastery and progress information.
GET	/api/chatbot/history/{student_id}	Retrieve previous chatbot interactions.
GET	/api/chatbot/past-paper/questions	Retrieve available past-paper practice questions.
POST	/api/chatbot/past-paper/evaluate	Evaluate a submitted practice answer.
GET	/api/chatbot/flashcards/{topic_id}	Retrieve topic-specific flashcards.
POST	/api/chatbot/flashcards/review	Record a flashcard review and update its review schedule.

POST	/api/chatbot/mock-exam/start	Start a mock examination session.
POST	/api/chatbot/mock-exam/submit	Submit answers and receive results and revision guidance.
GET	/api/chatbot/analytics/teacher	Retrieve teacher-facing learning analytics.
GET	/api/chatbot/repeated-alerts	Retrieve repeated-query alerts for teacher review.

The main chat response includes sourceType, fallbackReason and confidence alongside the answer. These fields allow the frontend to distinguish LLM-generated responses from local fallback responses. Student-specific endpoints require authorisation checks before accessing conversation history, progress records or recommendations.

4.5 Standards and professional practice

The component design considers accessibility, API security and responsible AI practices. Such guidelines affect the course of development without establishing proven compliance.

Accessibility

WCAG 2.2 offers guidance on text alternatives, contrast ratios, keyboard accessibility and focus indicators [10]. The chatbot design implements such considerations through accessible text content, control elements and explanations that remain functional beyond auditory delivery. Images, diagrams and other visual elements require descriptive labels, while student-facing explanations can be personalized according to their preferred modes of presentation.

API security and learner privacy

OWASP API Security Top 10 guides the management of authentication tokens, permission rules and resource pathways [11]. The design practice isolates LLM access credentials at the backend layer, while student-specific data and teacher-facing analytics require special permissions to access. The evaluation procedures anonymize participant data by design, with webcam feed recordings being excluded from external LLM prompts as part of the process.

Responsible AI

Learning state indicators are used as estimations rather than assessments of student comprehension. Response source indicators differentiate between LLM responses and local fallback solutions. Automated marking, progress and revision recommendations are presented as suggestions, while research assessments explicitly differentiate between intended and completed actions.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The Emotion-Aware Adaptive Chatbot was built using React as a front-end framework, FastAPI as a back-end framework, and MongoDB as a database to store the results of the learning process. The separation of the front-end, back-end, and data storage components allows for easier maintenance and further expansion of the application's features within the existing ecosystem.

Table 5.1: Implementation technologies and their purposes

Technology	Purpose
React and Vite	Develop and build the chatbot interface and related pages.
Tailwind CSS	Style the interface and support responsive layouts.
Axios and React Router	Handle API requests and navigation between pages.
Python and FastAPI	Provide backend endpoints and chatbot-processing services.
MongoDB	Store conversations, learning records and repeated-query alerts.
Configurable LLM service	Generate explanations using curriculum and learner context.
Local curriculum resources	Supply syllabus content and dataset-based fallback answers.

The frontend of the application consists of a chatbot, lesson summary, teacher analytics, and repeated query sections. The backend contains API routes, request/response models, adaptation services, and resource-ingestion mechanisms. Learning material comprises syllabus JSON, lessons, notes, and question banks organized into Grade 10 and Grade 11, including three semesters every.

Environment variables of the backend determine the LLM provider, model, and timeout. API keys are kept on the server and not shared in the research materials. A local dataset is used as an alternative response source if the target LLM is unavailable.

Software versions and test environments are omitted, restricting this experiment's replication.

5.2 Component implementation

The adaptive chatbot implementation involves curriculum selection, learner context, responses adaptation and answer generation. Students choose the relevant 10th or 11th-grade semester and topic to select the lesson context. Their questions are considered alongside available student feedback, conversation history, quiz performance and webcam-derived indicators.

The EARA engine adapts explanation style, length and difficulty. Students having difficulty get simpler explanations, prerequisite reminders or visual support. Bored students get diagrams, images and interactive examples, distracted students get shorter guidance, and students who understand get extension questions.

The backend configuration links the selected LLM to the local O/L ICT resources. If the LLM component raises an error or generates an invalid response, the local data is used as a fallback option. The response meta-information analyzes the selected solution and identifies the source of information and potential issues. The response includes relevant diagrams and images with descriptive text, if available.

The frontend displays the answer, along with annotations, practice problems, and sign-preview options. The conversation history contributes to the learner's context, progress tracking, and repeated question analysis. Figure 5.1. presents the chatbot's interface, with learning context selection, topic, and revision options.

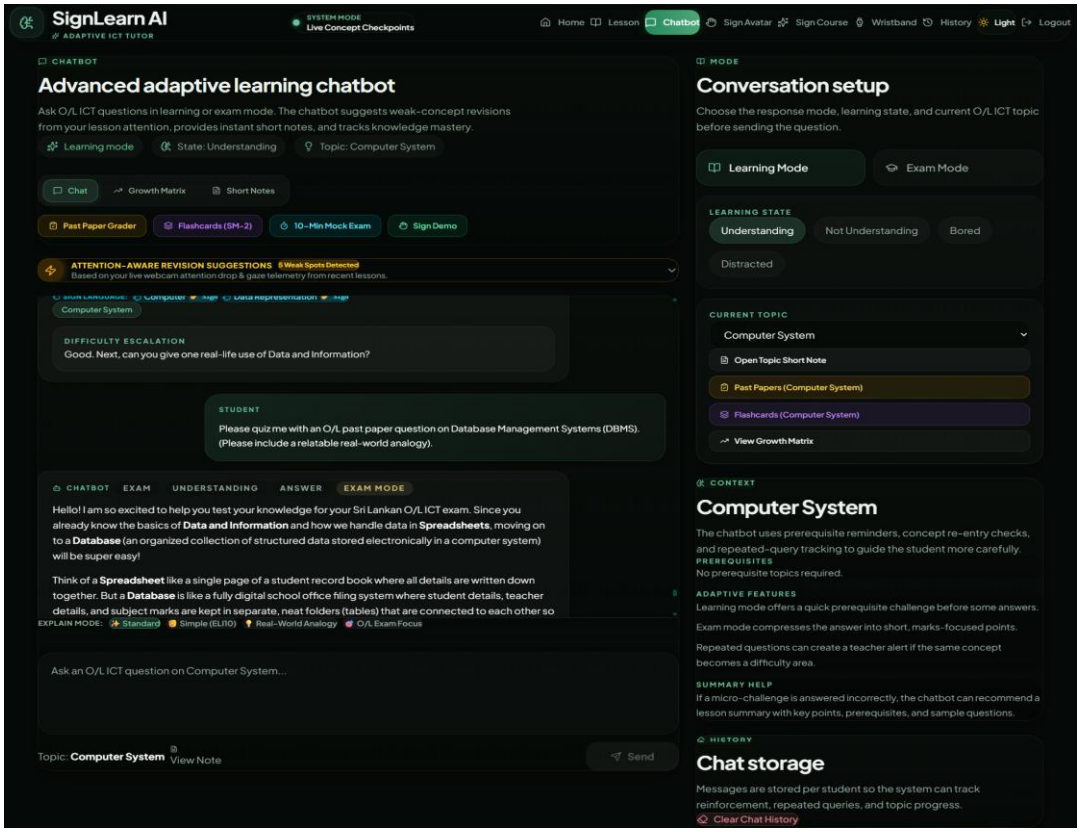


Figure 5.1: Adaptive chatbot interface showing learning-state selection, topic context and revision support

5.3 Integration with the other components

The adaptive chatbot exploits information and services provided by the attention-monitoring, knowledge-graph and sign-avatar components. These links bind chatbot assistance to the learner's broader learning activity.

Table 5.2: Integration with other platform components

Component	Information or service used	Purpose
Attention monitoring	Attention indicators and lesson-segment records	Suggest revision of potentially missed concepts.
Knowledge graph	Topic relationships and prerequisite concepts	Support prerequisite reminders and related practice.
Sign-avatar component	Sign previews for supported ICT terms	Provide additional visual explanations within chat.

Attention-based recommendations allow students to open a related short note or revise a concept using the chatbot. Knowledge-graph relations give prerequisite context and sign-preview affordances provide access to supported ICT signs. Student and topic identifiers tie these activities to the appropriate learner and lesson.

Integration tests will examine correct student and topic binding, revision facilities and management of unavailable data. Absent attention records are not to be taken as signs of understanding, unavailable sign previews should not render text explanation unusable. The chatbot's task is to exploit these services for learning support; development of underlying detection and sign-generation models is the responsibility of other components.

5.4 Verification and software testing

Functional testing was applied to certain features of the chatbot under investigation to check their general behavior. The developer stated that all 12 test cases provided in Table 5.3 were successful. These test cases included such areas as curriculum selection, response adaptation, revision activities, conversation retrieval, integration, and LLM failure handling.

Table 5.3: Functional test cases and observed outcomes

Test case	Expected behavior	Observed outcome	Status
Curriculum selection	Use the selected grade, semester and topic as response context.	Selected curriculum context was applied.	Pass
Valid ICT question	Return an answer relevant to the question and curriculum resources.	A relevant ICT answer was returned.	Pass
Not-understanding state	Provide simpler guidance, prerequisite support or a relevant visual explanation.	The response provided the expected learning support.	Pass
Bored state	Provide an engaging example, visual explanation or interactive challenge.	The response included the expected engagement support.	Pass
Explanation-style change	Apply the selected style to the subsequent response.	The subsequent response followed the selected style.	Pass
Exam Mode selection	Provide focused examination-oriented guidance.	Examination-focused guidance was displayed.	Pass
LLM service failure	Return a local answer with source and failure metadata.	Fallback returned an answer with the expected metadata.	Pass
Short-note request	Display notes for the selected topic.	The corresponding topic notes were displayed.	Pass
Micro-challenge submission	Check the answer and return feedback.	The submitted answer was checked and feedback returned.	Pass
Conversation retrieval	Retrieve messages for the correct student.	The relevant student's conversation was retrieved.	Pass
Attention recommendation	Open revision support for the associated concept.	The recommendation opened the relevant revision resource.	Pass
Repeated clarification	Generate an alert when the configured condition is met.	An alert was generated after the condition was met.	Pass

The indicated success rate for the function pass was 100% (12/12). These cases comprise 40% of the 30-case functional assessment that was planned. The results for the remaining 18 cases were not reported but neither failed nor passed.

The table only provides summary figures, without the actual request or response items, extracts, dates of execution, and references to relevant screenshots or logs. Therefore, the findings should be regarded as what the developers claimed, rather than independently verified claims.

The general evaluation that is described in chapter 3 has 96 controlled adaptation scenarios and 60 timed requests, but their scoring results and temporal records are not available. In turn, the functional success rate on its own does not confirm the response accuracy, adaptation adequacy for the targeted learners, or pedagogical effectiveness.

5.5 Deployment and operation

The chatbot is implemented in the shared ecosystem, with React frontend, FastAPI backend, and MongoDB with the stored learning records. The backend requires database connection, access to curriculum resources, and proper configuration of the LLM.

The frontend serves Vite scripts for developing, building, and previewing the project. The backend.env file defines the LLM type and model, API key, and timeout. The credentials should be kept private and never committed to the shared repository.

In the production state, the chatbot relies on the selected LLM as the primary source of information and generates answers based on the data retrieved from it. If the LLM becomes unavailable, the chatbot will utilize the data of the local set of documents to answer the question, if available. It should be noted that the fallback dataset only serves as a backup in case of LLM failure and does not replace the primary solution.

The operation of the chatbot should be supported through the monitoring of the API, database, curriculum, and fallback dataset availability. The maintenance will include the review of generated errors, updates to the lesson resources, and verification of topics across grades and semesters. In addition, changes to the adaptation rules or LLM utilization should trigger additional testing of the affected features.

6 RESULTS AND VALIDATION

6.1 Research results

Functional testing of 12 chatbot workflows was performed, and all test cases were successful. The tests were grouped by several criteria, including curriculum choice, response adaptation, revision activities, conversation retrieval, selected integrations and local fallback. Table 6.1 provides an overview of the test scope.

Table 6.1: Evaluation status at draft submission

Evaluation area	Available evidence or status
Functional testing	Twelve of the planned 30 cases were completed.
Controlled adaptation evaluation	The evaluation design covers 96 scenarios across curriculum contexts, learning states and interaction modes.
Response-time evaluation	The test design includes 60 requests across three response paths.
Student evaluation	The evaluation design considers navigation, explanation clarity and revision usefulness.
Teacher evaluation	The evaluation design considers accuracy, curriculum relevance and instructional suitability.

The completed cases were grouped into six functional areas, as shown in Table 6.2. Individual test cases and their observed outcomes are presented in Section 5.4.

Table 6.2: Summary of completed functional testing

Evaluation area	Cases tested	Passed	Failed
Curriculum selection and question answering	2	2	0
Learning-state adaptation, explanation styles and Exam Mode	4	4	0
LLM failure and local fallback	1	1	0
Short notes and micro-challenges	2	2	0
Conversation retrieval	1	1	0
Attention recommendations and repeated-query alerts	2	2	0
Total	12	12	0

Figure 6.1 illustrates the passed and failed cases by functional area. Response adaptation accounts for four cases covering misunderstanding, boredom, explanation-style changes and Exam Mode. Bar heights represent the number of cases tested rather than differences in feature quality.

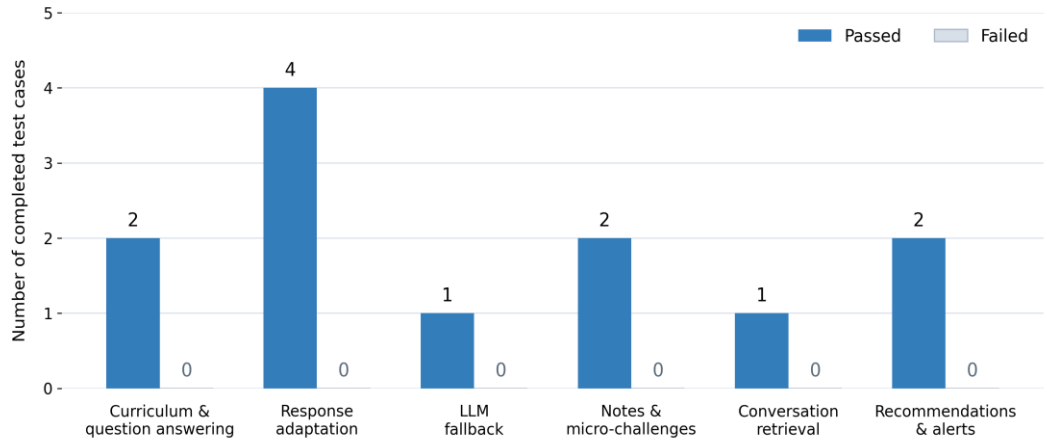


Figure 6.1: Passed and failed functional test cases by evaluation area

Table 6.3: Functional evaluation metrics

Metric	Result
Planned functional test cases	30
Completed and documented cases	12
Passed cases	12
Failed cases among completed tests	0
Pass rate among completed tests	100%
Proportion of planned cases documented	40%
Cases awaiting documented results	18

The outcome of the tests demonstrated that the chosen curriculum context was applied, and the response controls elicited the desired adaptations. The short notes were opened for the chosen topic, micro-challenges gave feedback, and conversation retrieval displayed the corresponding student's messages. Attention recommendations and repeated query alerts also worked as expected.

The LLM failure test triggered the local fallback and provided an answer with source metadata. This outcome confirms continuity in the context of the given failure scenario, though it cannot be considered reliable for all types of service interruptions.

The results confirm that the tested chatbot procedures worked as designed. The 100% passing rate was achieved across the 12 test cases executed. The functional tests, controlled adaptation assessment, response-time evaluations, and stakeholder assessments represent only a part of the overall scope; thus, the conclusion about the quality of answers and education efficiency is also limited to these aspects.

6.2 Product and technical results

The implemented chatbot puts together curriculum selection, adaptive explanations and revision support in a common interface. Its functionality was tested on the selected workflows for student input to response delivery. Table 6.4 below summarizes the results.

Table 6.4: Product and technical outcomes

Area	Outcome in completed testing
Curriculum context	The selected grade, semester and topic were applied to the interaction.
Response adaptation	Tested learning states, explanation styles and Exam Mode produced the expected response changes.
Revision support	Topic notes opened, and micro-challenge submissions returned feedback.
Conversation storage	Previous messages were retrieved for the relevant student.
Integration	Attention recommendations opened relevant resources, and repeated clarification triggered an alert.
Service continuity	The tested LLM failure activated the local fallback and returned source metadata.

These results prove that the tested features function during the chatbot's operation. The explanation controls let the students request different forms of assistance, while the revision links connected the interaction to further activities.

The fallback result proves an uninterrupted response during the failed procedure. It does not confirm each local answer's accuracy or invulnerability to all service flaws. The response's speed, throughput, and scalability were not tested and thus are beyond the scope of this technical analysis.

6.3 User or stakeholder validation

Stakeholder validation was included into the evaluation design to check if the chatbot would effectively respond to the needs of deaf and hard-of-hearing O/L ICT students and their teachers. Table 6.5 below contains the evaluation areas addressed by each stakeholder.

Table 6.5: Stakeholder validation focus

Stakeholder	Validation focus
Deaf and hard-of-hearing O/L ICT students	Ease of navigation, explanation clarity, usefulness of diagrams and images, and suitability of revision activities.
ICT teachers	Subject accuracy, curriculum alignment, appropriateness of adapted explanations, and usefulness of progress information and alerts.

The student evaluation procedure comprises choosing a topic, asking questions, altering explanation methods and completing a revision task. The teacher's review procedure focuses on evaluating the provided answers and further recommendations. Five-point ratings together with open-ended comments allow for addressing evaluation criteria efficiently. The completed evaluation was only a functional test conducted by developers. There is another step of student and teacher validation remaining; thus, it is still early to conclude that the chatbot has been completely accepted and positively used in classrooms.

6.4 Integration results

The completed tests offer initial evidence of integration between the chatbot, learning resources and supporting services. Attention-based recommendations successfully opened the associated revision resources, stored conversations were successfully retrieved for the correct student and repeated clarification raised an alert when the configured condition was reached.

The tested LLM failure correctly triggered the local fallback and returned the expected source metadata. This confirmed that an answer could be given in the case of this failure condition.

These results relate to the specific cases outlined in Section 5.4. Independent results for knowledge-graph prerequisite mapping, sign-avatar previews and operation when connected components are unavailable are not yet documented. The current results therefore represent selected integration workflows rather than full validation of the platform.

6.5 Summary of findings

- All 12 completed functional cases passed, representing 40% of the planned 30-case test set.
- Curriculum selection, learning-state adaptation and explanation controls behaved as expected in the tested interactions.
- Revision activities, conversation retrieval, attention recommendations and repeated-query alerts supported the tested workflows.
- The simulated LLM failure activated the local fallback and returned an answer with source metadata.

These findings support the chatbot's functional feasibility under the tested conditions. Broader adaptation testing, response-time measurements and student and teacher evaluation remain necessary to assess response quality, performance and educational usefulness.

7 DISCUSSION

7.1 Interpretation of findings

The completed evaluation considers functioning to have been achieved in distinction from learning effectiveness. All 12 carried-out cases passed, which constitutes 40% of the intended functional test set, supporting operations of the tested components but not consistent answering quality across the curriculum or improved comprehension.

The adaptation checks demonstrate that the learner's context could affect the presentation of assistance. Tested learning states, explanation styles and Exam Mode yielded the expected response alterations. However, differing wording or formatting does not guarantee that the explanation satisfies the student's point of need: the planned teacher ratings and student feedback are therefore important for evaluating fitness of such adaptations.

The fallback result demonstrates that the chatbot can sustain its responses during the tested LLM failure. This supports continuity of service, although availability and quality of answers requires separate interpretation. A local answer could be technically valid but offer limited support if the available resources do not fully address the query.

From a learning-support perspective, the component links explanations with revision activities and teacher follow-up. Attention recommendations aim learners at relevant resources, and repeated-query alerts identify common areas of request. These functions offer opportunity for support, but require effective topic matching and interpretation to be fit for purpose. Similarly, attention changes and repeated questions should prompt further assistance rather than direct adjudication of a student's understanding.

7.2 Comparison with literature baselines and existing solutions

Affective AutoTutor shows how dialogue tutoring can react to learners' cognitive and emotional states [5]. The chatbot then implements a related principle with EARA, but with focus on Sri Lankan O/L ICT contents and accessible explanation options. The performed tests then show selected adapted behaviors, but without proving similar effectiveness to the learning results described in prior research.

Retrieval-Augmented Generation uses retrieved information along with language generation [6]. The chatbot's local fallback then is intended for different reasons, as it maintains support when the chosen LLM fails. A successful fallback test thus rather proves continuity of service than benefits of applying RAG.

Moodle offers conditional learning pathways with completion and access rules [7], as does Hand Talk provide sign-language assistance for website contents [8]. The project then brings response adaptation, curriculum-based revision and access to supported sign previews together in a single chatbot workflow. Combined features however do not prove accessibility or usability advantages without direct evaluation.

The evaluation then does not include a controlled comparison to a non-adaptive chatbot or an LLM-only configuration. The current contribution thus only lies in its

implementation and initial verification. Comparing answer qualities, engagement and learning effectiveness would then require further testing.

7.3 Research contribution

The main contribution of the research involves designing and evaluating an adaptation process for a chatbot that responds to the context of interaction, including the curriculum-specific support for deaf and hard of hearing O/L ICT students. The contribution is manifested in the coordination of the discussed elements, not the creation of a novel language model or the development of algorithms for emotion detection.

EARA utilizes available learning-state indicators, student feedback, dialogue logs and quiz results to adapt explanations and organize the following activities. The grade and semester information serve to provide the context of the curriculum, while the diagram-based and image-rich explanations facilitate the students' understanding of the topic. In turn, the component is organized to establish connections between explanations and revision recommendations, default response and teacher alert triggers, thus shaping the process in which one question can demand clarification, practice, or teacher assistance.

The 12 implemented functional tests confirm the correct functioning of chosen adaptation steps, which in turn facilitate the proposed process execution. Therefore, the research has contributed to the solution by designing an adaptable process for a deaf and hard of hearing student to ask for clarification and receive revised explanations, which will be followed by the identification of the approach's effectiveness.

7.4 Product contribution

The product contribution is a chatbot component that delivers O/L ICT explanations, revision activities and progress information through the shared learning platform. Grade/semester organization enables students to access relevant content while learning modes and explanation controls let them request various kinds of assistance.

The chatbot combines micro-challenges, flashcards and practice assessments with text and visual explanations. Teacher analytics and repeated-query alerts inform follow-up support. Backend LLM integration and a local dataset fallback offer alternate response pathways if the selected service becomes unavailable.

The contribution delivers a base for expanded classroom experimentation and development. Its practical value depends on curriculum quality, dependable execution and feedback from the target student and teacher population.

7.5 Limitations and threats to validity

The completed evaluation included twelve functional cases. These checks represent specific behaviors but do not examine the curriculum in its entirety, account for all failure conditions or concurrent uses. Developer testing also includes potential confirmation bias, necessitating independent review.

Controlled learning-state labels can test the EARA's response rules, but not the accuracy of the webcam-derived indicators. Similarly, a response that follows an adaptation rule may still be ambiguous or inappropriate for the student. Functional pass rates therefore cannot be used as a substitute for assessment of answer quality or learning effectiveness.

LLM responses may vary between requests, while local fallback quality is dependent on dataset coverage and topic mapping. Model configuration, network conditions and timeout settings may also impact performance and consistency.

Student and teacher validation was outside the scope of the completed evaluation, limiting conclusions on accessibility and classroom usefulness. Without a baseline or long-term learning assessment, increases in understanding, engagement or retention cannot be attributed to the chatbot.

7.6 Implications

The implementation suggests that adaptive learning support should link student feedback, curriculum context and opportunities for revision. Students should have access to request an alternative explanation if the given answer is inappropriate or unsatisfactory. Diagrammatic and image-based answers provide additional presentation formats, although these may need to be evaluated in terms of relevance to the intended users.

For teachers, repeated-query alerts and progress information could inform future exploration of learning difficulties. These should be used to augment rather than replace classroom observations about a student's capacity. Likewise, shifts in attention should trigger potential learning supports rather than automatically implying misunderstanding.

Figure 7.1 depicts the interaction loop. Student responses update the context for subsequent support as well as informing teacher follow-ups to recurring difficulties. Note that the diagram reflects the intention of the support rather than asserting improvement in learning outcomes.

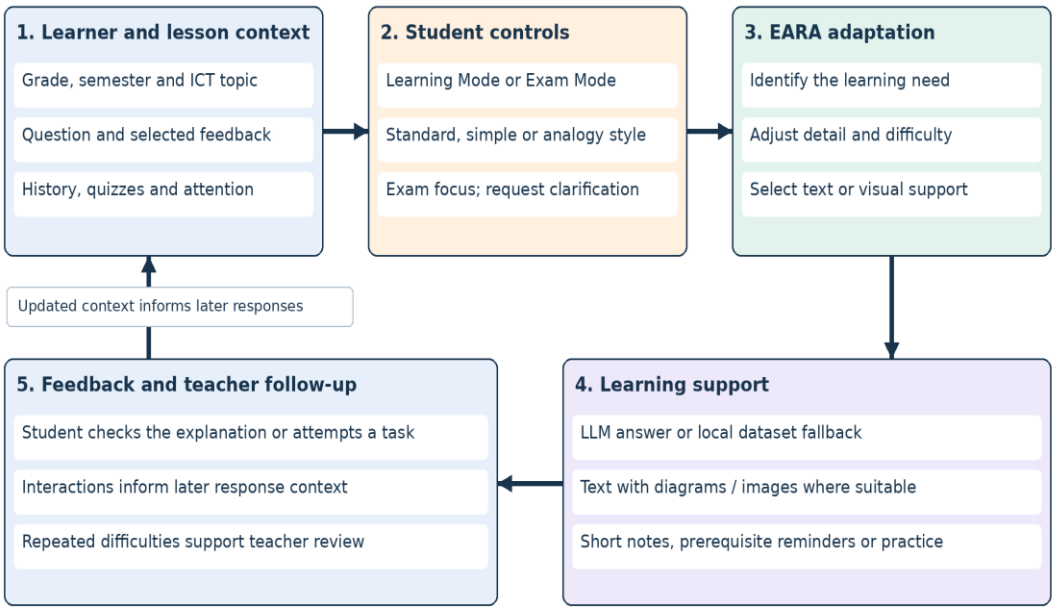


Figure 7.1: Adaptive learning support and teacher follow-up loop

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Target Users and Value Proposition

The target users of the proposed solution are deaf and hard-of-hearing students who study Sri Lankan O/L ICT and the teachers who provide support to them. The possible institutional adopters of the solution are the schools, special education centers, and organizations that offer access to knowledge.

The primary value of the proposed chatbot is the easy access to the necessary explanations and revision notes in terms of the curriculum. In addition, the organization of the notes by grade and semester helps the student find the necessary information quickly and easily. Within the scope of the proposed solution, students will also be able to choose the explanation method and the provision of visual information and graphs. The monitoring of teacher statistics and the analysis of frequently asked questions will provide valuable data for further assistance.

The solution can be adopted by schools through a pilot project to evaluate its effectiveness and practicality. Subsequently, based on the results of the pilot project, the solution can be provided for further licensing and adoption by institutions. At present, the commercial value, cost, and benefits for users have not been determined; therefore, the suggested plan is only a draft.

8.2 Commercialization Strategy

Commercialization might start with a limited institutional pilot using schools or educational organizations that work with deaf and hard of hearing students. This pilot can help identify the most appropriate classroom settings, workload, technical requirements and teacher readiness.

A possible business model is an institutional subscription covering platform access, maintenance, curriculum updates and user support. Grant funding or educational partnerships could help reduce costs for schools with limited resources. Pricing should be determined after measuring hosting, LLM usage and support costs.

The chatbot would be distributed as part of the institutional learning platform. Essential preparations would include additional user testing, ensuring the solution's accessibility and data security features, and determining responsibility for content maintenance. This approach would circumvent unproven assertions regarding enhanced learning outcomes while still gathering necessary data for institutional adoption.

8.3 Sustainability and Maintenance

Long-term functionality requires reliable hosting, cost management for the LLM, and maintenance of learning resources. Monitoring usage and having configurable limits

might help manage costs. The local fall-back is suitable for LLM failures, though it is dependent on the backend and available material.

Teachers or subject reviewers should evaluate lessons, question banks, and visual explanations for accuracy and syllabus alignment. Materials should preserve grade, semester, and topic categorizations when updated.

Updates, database maintenance, permissions auditing, and regression testing should be part of technical upkeep. Documentation and role definition would facilitate transition beyond the initial research team. Student and teacher feedback should be used to iterate and improve based on target user needs.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This research introduces the Emotion-Aware Adaptive Chatbot to facilitate deaf and hard-of-hearing students in understanding Sri Lankan O/L ICT course material. The solution effectively addresses the identified need by linking relevant lessons and personal learner data to explanations, revising activities and teacher-augmented features. Lessons are structured by grade level and semester while response adaptation includes streamlined guidance, visual explanations and follow-up practice sessions.

Implementation uses configurable large language model integration with local data-set fallback, while students select explanation styles and interaction modes. Conversation records, attention suggestions and repeated-query alerts contribute to subsequent assistance.

All 12 completed function cases passed, providing initial validation that the tested procedures perform as designed. These tests represent part of the proposed evaluation battery, while the study illustrates an implemented adaptive learning support approach, requiring further technical and stakeholder assessment to establish response quality, accessibility and pedagogical utility.

9.2 Achievement of objectives

The implementation addresses the main development objectives, while evaluation remains partially complete. Table 9.1 summarizes the current position.

Table 9.1: Achievement of research objectives

Objective	Achievement and evidence
Organize curriculum resources	Grade and semester organization was implemented, with curriculum selection checked in functional testing. Full content coverage requires review.
Combine learner information	The design incorporates student feedback, webcam-derived indicators, conversation history and quiz performance. Their combined effectiveness remains to be evaluated.
Develop EARA adaptation	Selected learning-state responses, explanation styles and Exam Mode passed the reported functional checks.
Integrate LLM and local fallback	The tested LLM failure produced a local answer with the expected metadata.
Provide targeted	Short notes, micro-challenges and attention recommendations passed the reported checks. Other

revision	revision features require further testing.
Support teacher intervention	Repeated-query alert behavior was checked. The usefulness of alerts and progress analytics requires teacher evaluation.
Evaluate the component	Twelve functional cases were completed and passed. Controlled adaptation assessment, response-time measurement and stakeholder validation remain further evaluation activities.

The development objectives were addressed through implementation, with selected behaviors checked through functional testing. The evaluation objective is partially achieved. Establishing educational usefulness requires completion of the remaining technical tests and direct evaluation with students and teachers.

9.3 Individual research and product contributions

A contribution to the project that I am most proud of is the design of EARA and its preliminary assessment. This feature links learner information and O/L ICT curriculum context with explanation strategies and revision activities, taking into account the students' feedback, interaction history, quiz, and webcam indicators, not as binary indicators of comprehension, but as soft factors.

My product contribution entails the chatbot interface, backend reaction services, curricular-resource integration and local fallback. The visual explanations, revision options, progress indication, and repeated query alerts were also designed. The relevant services designed by the other team members were utilized in the component.

The 12 functional tests executed qualify as evidence for this contribution. More assessments are required to evaluate the ability of the implemented feature to satisfy the needs of learners and educators.

9.4 Future directions

The future development should be concerned with finishing the evaluation and refining the chatbot according to the feedback from students and teachers. The major goals include:

- Complete the remaining functional tests, controlled adaptation scenarios and response-time measurements.
- Evaluate explanation clarity, visual accessibility and revision usefulness with deaf and hard-of-hearing students and ICT teachers.

- Compare EARA with a non-adaptive chatbot using the same curriculum resources and questions.
- Improve handling of missing or conflicting learner indicators while preserving student control over explanations.
- Expand and review curriculum materials, diagrams and supported sign previews across all grade–semester groups.
- Investigate longer-term understanding and retention through comparative learning assessments.

The above mentioned activities will help accumulate sufficient evidence for the educational effectiveness and make further decisions on classroom deployment.

REFERENCES

[1] C. W. Okonkwo and A. Ade-Ibijola, “Chatbots applications in education: A systematic review,” *Computers and Education: Artificial Intelligence*, vol. 2, Art. no. 100033, 2021, doi: 10.1016/j.caeai.2021.100033.

[2] R. E. Mayer, “Cognitive theory of multimedia learning,” in *The Cambridge Handbook of Multimedia Learning*, 3rd ed., R. E. Mayer and L. Fiorella, Eds. Cambridge, U.K.: Cambridge University Press, 2021, pp. 57–72, doi: 10.1017/9781108894333.008.

[3] S. D’Mello, B. Lehman, R. Pekrun, and A. Graesser, “Confusion can be beneficial for learning,” *Learning and Instruction*, vol. 29, pp. 153–170, 2014, doi: 10.1016/j.learninstruc.2012.05.003.

[4] J. D. Karpicke and H. L. Roediger III, “The critical importance of retrieval for learning,” *Science*, vol. 319, no. 5865, pp. 966–968, 2008, doi: 10.1126/science.1152408.

[5] S. D’Mello and A. Graesser, “AutoTutor and affective AutoTutor: Learning by talking with cognitively and emotionally intelligent computers that talk back,” *ACM Transactions on Interactive Intelligent Systems*, vol. 2, no. 4, Art. no. 23, 2012, doi: 10.1145/2395123.2395128.

[6] P. Lewis *et al.*, “Retrieval-augmented generation for knowledge-intensive NLP tasks,” *arXiv preprint arXiv:2005.11401*, 2020, doi: 10.48550/arXiv.2005.11401.

[7] Moodle, “Restrict access settings,” *MoodleDocs*. Accessed: Oct. 10, 2026. [Online]. Available: https://docs.moodle.org/502/en/Restrict_access_settings

[8] Hand Talk, “Parameters,” *Hand Talk Plugin Documentation*. Accessed: Oct. 10, 2026. [Online]. Available: https://docs.handtalk.me/en/docs/referencias-tecnicas/parametros/

[9] K. Peffers, T. Tuunanen, M. A. Rothenberger, and S. Chatterjee, “A design science research methodology for information systems research,” *Journal of*

Management Information Systems, vol. 24, no. 3, pp. 45–77, 2007, doi: 10.2753/MIS0742-1222240302.

[10] World Wide Web Consortium, "Web Content Accessibility Guidelines (WCAG) 2.2," W3C Recommendation, Dec. 12, 2024. Accessed: Oct. 10, 2026. [Online]. Available: https://www.w3.org/TR/WCAG22/

[11] OWASP Foundation, "OWASP Top 10 API Security Risks—2023," 2023. Accessed: Oct. 10, 2026. [Online]. Available: https://owasp.org/API-Security/editions/2023/en/0x11-t10/

GLOSSARY

Term	Definition
Adaptive response	An explanation whose style, length, difficulty or follow-up support changes according to the learner's context.
Attention-based recommendation	A suggestion to revisit lesson content associated with a possible drop in attention.
Curriculum grounding	Using relevant syllabus and lesson resources to guide the subject content of an answer.
Knowledge graph	A structured representation of concepts and their relationships, including prerequisite links.
Learner context	Available information about a student's question, selected feedback, previous interactions, assessment performance and learning situation.
Learning-state indicator	A signal used to estimate the support a learner may need, rather than a confirmed measurement of emotion or understanding.
Local fallback	An alternative response path that uses locally available curriculum resources when the configured language model is unavailable or fails.
Mastery score	A system-generated indicator of topic understanding based on recorded learning activity; it is not a validated examination prediction.
Micro-challenge	A brief question or task used to check understanding of a concept or prerequisite.
Prerequisite concept	Earlier knowledge needed to understand a more advanced topic.
Repeated-query alert	A notification identifying recurring requests for clarification about a concept for teacher review.
Spaced repetition	Reviewing material at intervals to support longer-term recall.

APPENDICES

Appendix A: Evaluation Instruments

The evaluation described in Chapter 3 uses the following instruments:

Instrument	Information collected
Student questionnaire	Five-point ratings and comments on clarity, navigation, visual support and revision usefulness.
Teacher review form	Five-point ratings of accuracy, curriculum relevance, clarity and adaptation suitability.
Technical test record	Test input, expected and observed outcomes, pass/fail status, supporting evidence and response time where applicable.

These instruments support evidence collection. Completed records should be retained separately, with participant identities protected.

Appendix B: Academic Integrity and AI-Use Disclosure

OpenAI ChatGPT assisted with thesis organization, drafting, language revision, reference discovery and formatting, evaluation planning, diagrams and charts. Representative requests included shortening sections, explaining the architecture and presenting reported test outcomes.

Project descriptions were based on supplied documentation. Sources were checked through publisher or official websites during drafting, but the source code and backend were not independently verified. The author reported the 12 passed functional cases; the larger evaluation quantities remain planned.

AI-assisted material affects Chapters 1–9, supplementary sections and generated figures. The author is responsible for final verification, supporting evidence and compliance with the module's AI-use policy.